

Speech-To-Text conversion for Oral histories and interviews

Yoav Haviv Vaknin , Dimitri Shpak

Academic advisors: Mr idan Tubis , Prof Shlomo Grinberg

Motivation

Personal testimonies of soldiers from World War II are an invaluable historical source, but many of them are in old formats, in different languages, and are not accessible to the general public. Today, in the digital age, it is of immense importance to preserve this heritage and make it accessible to diverse audiences

Goal

The goal of the project is to develop a system that will be used by the Jewish World War II Museum to transcribe and summarize the testimonies of into various languages for the purpose of understanding the testimony and researching the history

The solution

Frontend (React.js)



Built using React and Material-UI for a responsive, accessible design.
Enables users to:

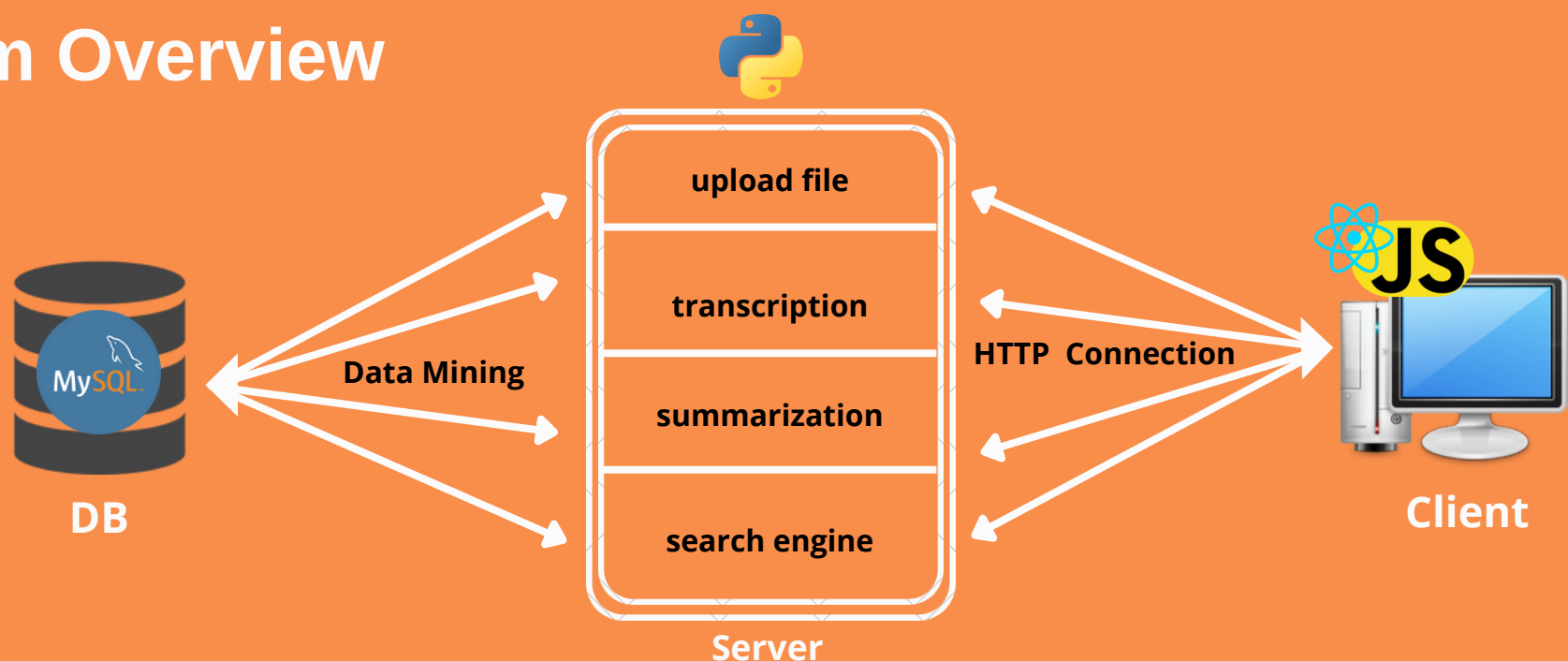
- Upload testimony videos
- Perform a keyword search in video summaries and titles
- Access to multilingual summaries and transcripts (Hebrew, English, Russian)
- Token-based authentication for secure access

Backend (Python Flask)



- RESTful API server built with Flask
- Endpoints for transcription, summarization, translation, and search
- Uses Faster-Whisper for speech-to-text transcription
- Uses NLTK for generating concise summaries
- Uses GoogleTranslator for multilingual translation (Hebrew, English, Russian)
- Modular design with separate logic for each NLP task

System Overview



Conclusion

This project delivers an end-to-end system for preserving and accessing WWII veterans' testimonies. It combines transcription, summarization, and translation powered by AI, and presents the results through a user-friendly React interface. The platform supports multi-language access and searchable content, ensuring these valuable historical records remain accessible for future generations.

For the code project
scan the QR :

